6TH GRADE MATH · GEOMETRY

Parallelogram & Triangle Area Lab

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Parallelogram & Triangle Area Lab** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. A rectangle is 6 by 4. You push the top sideways to make a parallelogram, keeping the same base and height. Does the area change?
   No. It stays 24 square units, because the base and height are the same.
2. A parallelogram has base 8, height 5 and slanted sides 6. What is its area?
   40 square units (base × height). The slanted side is not used.

2 Explore use the tool

1. Set base 6 and height 4. Lean the top to 3, then to −2. Record the area each time.
   24 square units both times.
2. With lean 3, press Show cut & slide. What shape do the pieces make, and what are its dimensions?
   A rectangle, 6 by 4.
3. Set base 5, height 4, lean 6. Compare base × slanted side with the real area.
   Area 20. The slanted side is √52 ≈ 7.21, so base × slanted side ≈ 36.1, which is wrong.
4. Switch to Triangle and show the copy. What shape do the triangle and its copy make? How does that explain the formula?
   A parallelogram with the same base and height. The triangle is half of it: $\frac{1}{2} \times b \times h$.

3 Record use the tool and fill in the table

Areas

Use the sliders. For the last row, find the missing base.

Shape	Base	Height	Lean	Area (square units)
Parallelogram	8	3	0	24
Parallelogram	8	3	5	24
Triangle	8	3	2	12
Triangle	8	3	−3	12
Triangle	6	4	any	12

4 Explain what did you discover?

1. Leaning a parallelogram does not change its **base** or its **height**, so the area stays the same.
2. Area of a parallelogram = **base × height** .

3. A triangle is **half** of a parallelogram with the same base and height, so its area = $\frac{1}{2}$ × base × height.

4. The height is measured **straight up (perpendicular)** from the base, even when it falls outside the shape.

5. Why doesn't the length of the slanted side matter for area?

   Area counts square units. Cutting off the slanted end and sliding it to the other side makes a rectangle with the same base and height. The slanted side gets longer as you lean, but the area stays the same.

5 Apply a new situation, no tool

1. Find the area of a parallelogram with base 12 cm, height 7 cm and slanted sides 9 cm.

   84 cm²

2. A triangle has an area of 30 in² and a base of 10 in. Find its height.

   6 in.

3. Two triangles have the same base and the same height, but one is obtuse. Compare their areas.

   They are equal.

4. A triangle's base is doubled and its height stays the same. What happens to the area?

   It doubles.

Teacher notes

Intended discovery: Area depends on base and perpendicular height only; shearing preserves both, so parallelograms rearrange into rectangles and triangles are half of a parallelogram.

Likely misconceptions:

- Multiplying base × slanted side (Explore 3 and Try this 3).
- Thinking an obtuse triangle has no height, or measuring the height inside the shape.
- Forgetting the $\frac{1}{2}$ for triangles.

Answers that can vary: Try this 2 and 5 accept many shapes (any b × h = 24 with a lean; any triangle with b × h = 24 and the top vertex past the base).